MikroTik NAT Port Forward for PULSE

Configuration, testing and troubleshooting record

Router	MikroTik RB2011UiAS (RouterOS v6.49.15), identity: Saka
Public IP	197.234.57.98 (WAN interface)
Target server	PULSE server, 192.168.155.50
Ports forwarded	TCP 59101 and TCP 59105 (same port on both sides)
Allowed sources	125.209.72.136/30 and 61.5.139.184/29 (vendor subnets)
Test source	AWS EC2 (Ubuntu), public IP 3.91.255.201 (temporary)
Date	20 October 2026

1. Summary

The vendor needs access to the PULSE server through the router's public IP, on TCP ports 59101 and 59105, and only from their two subnets. The forward was configured with an address list, a dst-nat rule and a forward accept rule. It was tested from a temporary AWS EC2 instance whose IP was added to the allowed list.

The first tests timed out. The cause was a single wrong field in the NAT rule: the router's public IP had been entered as the **Src. Address** instead of the **Dst. Address**. After correcting it, port 59101 returned *succeeded* from the EC2 instance.

2. Requirements

- Forward TCP 59101 and 59105 on 197.234.57.98 to 192.168.155.50 on the same ports.
- Restrict the forward to 125.209.72.136/30 and 61.5.139.184/29 only.
- Confirm it works before telling the vendor.

3. Final configuration

3.1 Address list (the allowed sources)

RouterOS cannot take two subnets in one Src. Address field, so both subnets are stored in one address list that the rules reference. *WinBox: IP > Firewall > Address Lists.*

List name	Address	Note
PULSE-ALLOWED	125.209.72.136/30	Vendor subnet 1
PULSE-ALLOWED	61.5.139.184/29	Vendor subnet 2
PULSE-ALLOWED	3.91.255.201	TEMPORARY - AWS test, removed after testing

```
/ip firewall address-list add list=PULSE-ALLOWED address=125.209.72.136/30
/ip firewall address-list add list=PULSE-ALLOWED address=61.5.139.184/29
```

3.2 dst-nat rule

WinBox: IP > Firewall > NAT. To Ports is left blank so the ports stay the same (59101 to 59101, 59105 to 59105). Entering a value there would remap both ports to it.

Field	Value
Chain	dstnat
Protocol	tcp
Dst. Address	197.234.57.98
Dst. Port	59101,59105
In. Interface	WAN
Src. Address List	PULSE-ALLOWED
Action	dst-nat
To Addresses	192.168.155.50
To Ports	(blank)
Comment	PULSE port forward - restricted

```
/ip firewall nat add chain=dstnat protocol=tcp dst-address=197.234.57.98 \
    dst-port=59101,59105 in-interface=WAN src-address-list=PULSE-ALLOWED \
    action=dst-nat to-addresses=192.168.155.50 \
    comment="PULSE port forward - restricted"
```

3.3 Forward accept rule

WinBox: IP > Firewall > Filter Rules. After dst-nat the forward chain sees the destination as 192.168.155.50, not the public IP, so the accept rule must match the internal address. It must sit above any drop rule in the forward chain.

Field	Value
Chain	forward
Protocol	tcp
Dst. Address	192.168.155.50
Dst. Port	59101,59105
Src. Address List	PULSE-ALLOWED
Action	accept
Comment	PULSE port forward - acceptance rule

```
/ip firewall filter add chain=forward protocol=tcp dst-address=192.168.155.50 \
    dst-port=59101,59105 src-address-list=PULSE-ALLOWED action=accept \
    place-before=0 comment="PULSE port forward - acceptance rule"
```

4. Test method

Because the vendor's source subnets cannot be tested directly, a temporary EC2 instance was used as a stand-in source.

Step	Action
1	Launched an Ubuntu t3.micro EC2 instance in a public subnet with auto-assigned public IP (test traffic must leave through the Internet Gateway, not the site-to-site VPN).
2	Confirmed the instance's outbound IP with `curl -4 https://checkip.amazonaws.com` (result 3.91.255.201).

Step	Action
3	Added 3.91.255.201 to the PULSE-ALLOWED list with the comment 'TEMP - AWS test, remove after'.
4	Tested with nc `-vz -w 10 197.234.57.98 59101` (and 59105).

5. What happened: troubleshooting timeline

#	Observation	What it told us
1	nc to 59101 and 59105 timed out. Ping to 197.234.57.98 worked (about 250 ms).	Router reachable; ICMP is allowed on input. Says nothing about the forward.
2	`curl -s ifconfig.me` printed nothing.	-s hid the error. Re-ran with checkip.amazonaws.com and got 3.91.255.201.
3	Address list entry 3.91.255.201 matched the EC2 IP exactly.	Source IP and list entry were correct. Problem was elsewhere.
4	Connections filter for 3.91.255.201 was empty.	Not conclusive: SYN entries expire in about 5 seconds and dropped packets are not listed.
5	Torch on WAN showed 3.91.255.201 to 197.234.57.98:59101, 1 Rx packet, 0 Tx.	Packets reach the router, so the ISP and the public address are fine.
6	Torch on LAN (dst 192.168.155.50) showed 0 packets.	Nothing was being forwarded to PULSE.
7	Filter rule 'PULSE acceptance' showed 0 bytes. Forward chain had no active final drop (the 'Drop All Other' rule was disabled). Raw table empty.	Packet never arrived at the forward chain addressed to 192.168.155.50. The dst-nat was not translating. (An earlier guess that the forward chain was dropping it was corrected here.)
8	`/ip firewall nat print detail` showed rule 0 with `src-address=197.234.57.98`.	**Root cause.** The rule only matched packets coming FROM the router's own IP.

5.1 Root cause

The NAT rule contained `src-address=197.234.57.98`. That is the router's public IP, so the rule matched only traffic whose source was 197.234.57.98. Real traffic from 3.91.255.201 (or from the vendor subnets) never matched, the packet stayed addressed to the router itself, and the connection timed out. The public IP belonged in **Dst. Address**, not **Src. Address**.

5.2 Fix applied

```
/ip firewall nat unset 0 src-address
/ip firewall nat set 0 dst-address=197.234.57.98
/ip firewall nat print detail where comment~"PULSE"
```

Resulting rule: chain=dstnat action=dst-nat to-addresses=192.168.155.50 protocol=tcp dst-address=197.234.57.98 src-address-list=PULSE-ALLOWED in-interface=WAN dst-port=59101,59105, with no src-address left on it.

6. Verification

Test	Result
nc -vz -w 10 197.234.57.98 59101 from EC2 (3.91.255.201)	**Connection succeeded** - NAT, forward accept and PULSE listener all working

Test	Result
nc -vz -w 10 197.234.57.98 59105 from EC2	Pending confirmation
Test from an IP not in PULSE-ALLOWED (for example a phone hotspot)	Pending confirmation. Expected result: timeout, proving the source restriction

How to read nc results: **succeeded** means the forward works end to end; **refused** means the forward works but nothing is listening or the server firewall rejected it; **timeout** means the packet was dropped (check rule order, the address list entry and the counters).

7. Cleanup before telling the vendor

Done	Task	Command
[ ]	Remove the temporary EC2 address-list entry	/ip firewall address-list remove [find comment~"TEMP - AWS test"]
[ ]	Confirm only the two vendor subnets remain	/ip firewall address-list print where list=PULSE-ALLOWED
[ ]	Remove any temporary log or test rules	/ip firewall raw remove [find comment=TEMP-LOG] /ip firewall filter remove [find comment=TEMP-LOG] /ip firewall nat remove [find comment=TEMP-TEST]
[ ]	Terminate the EC2 instance	AWS Console > EC2 > Instances
[ ]	Release any Elastic IP (unattached ones are billed)	AWS Console > EC2 > Elastic IPs

8. Lessons and notes

- Check the NAT rule fields first when a forward times out. A source address in place of a destination address is easy to miss and makes the rule silently never match.
- Torch on both WAN and LAN quickly shows whether packets arrive and whether they are forwarded.
- A rule counter of 0 on both the NAT rule and the forward accept rule points at the NAT rule not matching.
- Keep the PULSE NAT rule above the dynamic hotspot dstnat jump rules.
- Filter rule 'Allow Pulse Traffic to Router' (forward chain, 0 hits) appears to be a leftover from an earlier attempt; review it.
- Use a dedicated address list for allowed sources so adding or removing a subnet does not require editing the rules.
- If the server still does not answer with both counters moving: its default gateway must be the MikroTik, its host firewall must allow TCP 59101 and 59105 from outside the local subnet, and the service must listen on those ports.

9. Command quick reference

```
# Inspect
/ip firewall nat print detail without-paging
/ip firewall filter print stats without-paging where chain=forward
/ip firewall address-list print where list=PULSE-ALLOWED

# Reset and re-test
/ip firewall nat reset-counters-all
/ip firewall filter reset-counters-all
```

```
# Live view
/tool torch interface=WAN src-address=3.91.255.201/32 port=59101
/tool sniffer quick interface=LAN ip-address=192.168.155.50 port=59101
/ip firewall connection print where src-address~"3.91.255.201"
```